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Available Saildocs Grib Data

20-26 minutes

Subject: Available Saildocs Grib Data

"Grib" is short for "gridded binary", a data format originally used for computer forecast-model data but also used for forecaster-generated data (NDFD forecasts).

There are a number of forecast models available: The two leading global models are NOAA's GFS and the European ECMWF. GFS data is available, and limited ECMWF data is publicly available (see below). The ICON global model from Germany is also available, as is NAVGEM from the US Navy, and while not as highly regarded as GFS and ECMWF are useful for a "second opinion".

There are also a number of regional models available: Icon Europe, COAMPS and NAM for North America, and the hi-res HRRR forecast for US coastal waters (updated hourly). Wave data is available globally from the WW3 (GFSwave) model, and ocean current forecast data is available from NOAA's RTOFS model. OSCAR provides average currents for the previous 5 days, see notes below.

For general information about requesting grib data via email from Saildocs, send a blank email to: gribinfo@saildocs.com or see the web at <https://saildocs.com/gribinfo>

Computer Forecast Models in Grib format:

The primary source for Saildocs computer-model data is NOAA's "GFS" global model, requested by using the code

"GFS:" (or "Grib:") in a grib-request. Data from other models is also available. Here are the details for the available models:

General notes:

Forecast models are run periodically, and create forecasts from a zero-hour "synoptic time" (abbreviated as "HH"). NOAA's GFS model is run every 6 hours with synoptic times of 00z, 06z, 12z, 18z (UTC). Availability of the data is some hours after that: 4 to 5 hours for GFS, 8 hours for the European ECMWF model, a bit earlier for the new AI models. Obviously the earlier a forecast is available, the more useful it is. And remember, all times are UTC-- weather is a global phenomena.

GFS:

The most-often used model is NOAA/NCEP's Global Forecast System ("GFS") model. This model is run every 6 hours (HH=00, 06, 12, 18z) and starts becoming available at HH+03:40. Saildocs tracks the model run and makes data available as it is generated. Generally the first 72 hours forecast-hours are available by HH+04:00, 120 hours by HH+4:15, 240 hours HH+4:45, and complete by HH+5:30.

Remember that the accuracy of any model declines with increasing forecast times, and 5-7 days is usually considered a useful limit.

Scheduled requests (subscriptions) are sent when the requested forecast times become available. The shorter the forecast, the earlier you get it. So if you want a 72-hour forecast from the 00z model run, request time=06:00 and it will be sent around 04:00. Use the synoptic time plus 6 hours as the request time, for example time=06:00 for the 00z run, and it will be sent earlier whenever the requested forecasts become available. (Saildocs will send earlier, but not later

other than congestion).

Pro tip #1: 00:00z is the default time for text-page requests, and gets congested. And 06:00z is the default for grib and is also congested.

Pro tip #2: On a passage, consider a "low-beam/high-beam" approach: Two scheduled requests, one for 0-72 hours, another for 78-240 (or whatever your planning horizon is), time=06:00 (6 hours after synoptic time) and interval=6 (00, 06, 12, 18z synoptic time). The 3-day forecast will be sent around 4:10, the 10-day around 4:45, and every 6 hours.

This works for GFS, also HRRR and other NOAA moels (with different timing), but not for ECMWF and other moels which are dumped on the server as a batch.

Currently available GFS parameters are:

- PRMSL (or MSLP, Pressure reduced to Mean Sea Level)
- WIND (10 meters above surface)
- GUST (at 10 meters)
- TMP (temperature 2 meters above surface)
- SFCTMP (surface temperature)
- RH (Relative Humidity 2m above surface)
- LFTX (LiFTed indeX)
- CAPE (Clear Air Potential Energy)
- CIN (Convective INhibition)
- RAIN (Precip rate, mm/hr)
- APCP (Accumulated precip)
- REFC (Composite Reflectivity)
- HGT500 (500mb height)
- TMP500 (temperature at 500mb level)
- WIND500 (Wind velocity at 500mb level)
- HGT850 (850mb height)
- TMP850 (temperature at 850mb level)
- WIND850 (Wind velocity at 850mb level)
- ABSV (Absolute vorticity at 500mb)
- CLOUDS (Total cloud cover)

- ICEC (Ice Concentration)
- ICETK (Ice Thickness)

Wave data (GFS-wave, aka WW3) is now included with the GFS files-- see WW3 below.

Data is currently available as 0,1..24,27..192,198..384 forecasts, all on a 0.25 x 0.25-deg grid. In other words:

- 0-24 hours every hour
- 27-192 hours every 3 hours
- 198-384 hours every 6 hours

Note, however, that forecast accuracy for any model becomes increasingly speculative after 4-6 days.

A typical request could be:

```
send GFS:26N,20N,114W,105W|0.25,0.25|0,6..72|  
WIND,PRMSL,WAVES
```

Why do we list GFS first? Most weather-guru's rank global weather models as ECMWF first, GFS second, others behind those. That is based on analyses generally done long after the weather event being studied, and don't consider the effect of timing. We've listed GFS here first, not out of any loyalty, but simply because tomorrow's forecast is available from GFS a full 4 hours earlier. In the sailing world, where the unexpected is expected, that's the difference between the 6-hour chart being two hours in the future versus two hours in the past.

WW3 / GFSwave:

This is NOAA's WW3 wave model, run in conjunction with GFS on the same schedule and grid (see above) and renamed to GFS-wave and combined with GFS by Saildocs. All of the primary WW3 data is available, the default parameter is "HTSGW" (height of significant waves, i.e. the average of the largest 1/3 of combined swell and wind-waves), "WAVES" means the same.

GFS-wave parameters available from Saildocs are as follows:

- HTSGW Significant Height of Combined Wind Waves and Swell [100]
- DIRPW Direction of the Primary Wave (towards) [107]
- PERPW Period of the Primary Wave [108]
- WVHGT Wind Waves Significant Height [102]
- WVDIR Wind Waves Direction (towards) [109]
- WVPER Wind Waves Mean Period [110]
- SWELL Swell Wave Sig. Height (primary swell) [105]
- SWDIR Swell Wave Direction [104]
- SWPER Swell Wave Mean Period [106]

The "WW3" code can still be used for wave data, or "GFS" for both atmospheric and wave data. A typical request could be:

```
send GFS:26N,20N,114W,105W|0.25,0.25|0,6..72|  
WIND,HTSGW,DIRPW,PERPW
```

Note: The [...] codes are grib1 codes used for the various parameters. There is some ambiguity in WW3 grib codes particularly for WVDIR and WVPER.

GFSwave also provides spectral data for three swell patterns but this creates compatibility issues for Grib1 coding, so Saildocs currently provides only the primary swell as SWELL, SWDIR, SWPER. Separate codes may be added in the future using layer-type 241 "Ordered Sequence of Data".

Also note that NOAA changed the model-code (the "Generating Process ID" code, byte 6 of the PDS section) from 88 ("NOAA Wave Watch III (NWW3) Ocean Wave Model") to 11 ("Global Multi-Grid Wave Model (Static Grids)"). This makes no difference to the model data, but causes zyGrib software to not display wave data. To avoid this issue Saildocs substitutes the previous NWW3 code.

Regional WW3 models are now redundant with GFS-wave and are discontinued. The exception is the WW3 European regional model which includes the Black Sea, which the global

GFSwave/WW3 model does not:

ww3euro: bounded by 29n,66n,015w,045E, 0.2 x 0.2 grid, to 96 hrs @ 6 hr increments

AI GFS:

NOAA has released an AI-based GFS global model (AIGFS) for evaluation, and from Dec 17 2025 as an experimental release designated as AIGFS. AIGFS is a data-driven deterministic model based on Google DeepMind's GraphCast model. This model, and ECMWF's AI version of their IFS global model AIFS, share the initialization with their physics-based counterparts but use history rather than physics to predict future weather. It's a fundamentally different approach, and very much experimental at this point. Since initialization is shared with GFS the first few forecast days will be similar but then likely diverge.

This model is run every 6 hours (HH=00,06,12,18z) and starts becoming available at HH+03:40. Saildocs tracks the model run and makes data available as it is generated, generally finished out to 360 hours by HH+4:20.

Currently available AIGFS parameters are:

- PRMSL (PPressure/Mean Sea Level)
- WIND (10 meters above surface)
- TMP (temperature 2 meters above surface)
- SFCTMP (surface temperature)
- APCP (Accumulated precip)

Data is currently available for 6-hourly forecasts out to 384 (16 days), at this writing the zero-hour forecast (analysis) is not always available.

And do note that forecast accuracy for any model becomes increasingly speculative after 4-6 days.

More info (external links):

https://www.weather.gov/media/notification/pdf_2025/

[scn25-89_AIGFS_AIGEFS_and_HGEFS.pdf](#)

<https://deepmind.google/blog/graphcast-ai-model-for-faster-and-more-accurate-global-weather-forecasting/>

NAM:

NAM is NOAA's North Americal Mesoscale model for continetal US and surrounding waters. NAM is updated every 6 hours, and covers roughly 20N-60N, and 140W to 060W.

To see the extent of coverage, send this grib request to query@saildocs.com (about 40KB):

Send NAM:70N,00N,160W,050W|0.5|0|WIND

Currently available NAM parameters are:

- PRMSL (PResure/Mean Sea Level)
- WIND (10 meters above surface)
- GUST (at 10 meters)
- TMP (temperature 2 meters above surface)
- SFCTMP (surface temperature)
- RH (Relative Humidity 2m above surface)
- APCP (Accumulated precip)
- REFC (Composite Reflectivity)
- HGT500 (500mb height)
- TMP500 (temperature at 500mb level)
- WIND500 (Wind velocity at 500mb level)

NAM is updated every 6 hours, available by 4 hours after synoptic time (HH+4:00), and available forecast times are hourly to 36 hours, and 3-hourly to 84 hours (0,1..36,39..84 in Saildocs terms).

Note that NAM is scheduled to be discontinued by NOAA in early 2026, replaced by the RAP model below.

RAP:

RAP is NOAA's Rapid Refresh mesoscale (13-km) model,

updated hourly with forecasts at one-hour interval to 21 hours, and updated every 3 hours (00z, 03z, etc) with forecasts at one hour intervals to 51 hours. RAP data is available now, and is scheduled to replace NAM in early 2026.

Coverage is roughly 20N-60N, 130W-065W, and the lat-lon data grid is 0.10 deg.

To see the extent of coverage, send this grib request to query@saildocs.com (about 30KB):

RAP:70N,00N,160W,050W|0.5|0|WIND

Currently available RAP parameters are:

- PRMSL (PResure/Mean Sea Level)
- WIND (10 meters above surface)
- GUST (at 10 meters)
- TMP (temperature 2 meters above surface)
- SFCTMP (surface temperature)
- RAIN (Precip rate, mm/hr)
- APCP (Accumulated precip)
- REFC (Composite Reflectivity)
- CLOUDS (Total cloud cover)

HRRR:

NOAA's "High Resolution Rapid Refresh" is a hi-res forecast updated hourly, with forecast intervals as short as 15 minutes. HRRR primarily covers the Continental US but does spill into the coastal waters-- very roughly east of a line from 24N 123W to 48N 133W in the Pacific, and 23N 074W to 48N 063W in the Atlantic.

To see the extent of coverage, send this grib request to query@saildocs.com (about 18KB):

Send HRRR:55N,20N,140W,060W|0.5|0|WIND

Maximum resolution is 0.025 x 0.025 deg (around 2 miles) but care is needed: With large areas, high resolution and hourly

increments (or less) the files can easily get huge.

The hourly update is code HRRR and available out to 18 hours. Data is available at HH+01:30 where "HH" is the hourly synoptic time of 00z, 01z, etc. So for example the 00z update cycle is available at 01:30 utc. Saildocs will always send the most-recently available data. The model run is also part of the file-name listed in the reply-message.

Sub-hourly data is also available, 0.25-hour (15 min) forecast intervals out to 6 hours, 0.5 hour intervals to 12 hour and hourly beyond that. Note that Sub-hourly data from NOAA does not include SFCTMP, LFTX, CAPE, CIN.

There is also an extended model run (code HRRRX) every 6 hours which goes out to 48 hours, available by HH+02:00. Forecast interval for the first 18 hours is as above, then hourly.

Available HRRR parameters are:

- PRMSL (pressure at sea level)
- WIND (10 meters above surface)
- GUST (10 meters)
- AIRTMP or TMP (temperature 2 meters above surface)
- SFCTMP (temp at surface)
- LFTX (LiFTed indeX)
- CAPE (Clear Air Potential Energy)
- CIN (Convective INhibition)
- RAIN (Precip rate, mm/hr)
- CLOUDS (Total Cloud Cover)
- REFC (Composite Reflectivity)

A sample file could be this, sent to query@saildocs.com (about 35KB):

```
send HRRR:35N,31N,122W,116W|0.2,0.2|0,1..18|WIND
```

ECMWF

The European ECMWF IFS global model is highly regarded, but the full ECMWF catalog is not available for distribution without prohibitive fees. Previously, only very limited data was available as "WMO Essential" data to satisfy international obligations. However, ECMWF has now extended their "Open Data" project to include their primary "Integrated Forecast System" (IFS) model data (as well as AIFS, below), at reduced resolution and delayed from paid access. Saildocs now provides this data which can be freely distributed (see ECMWF terms below).

Currently available parameters are:

- WIND (10 meters above surface)
- PRMSL (PResure/Mean Sea Level)
- TMP (temperature 2 meters above surface)
- HGT500 (500mb height)
- HTSGW or WAVES) Significant Height of Combined Wind Waves and Swell)
- DIRPW (Direction of the Primary Wave - traveling towards)
- PERPW (Period of the Primary Wave)

The code for ECMWF's IFS data is "ECMWF", default parameters are surface pressure and wind. The grid is 0.25 x 0.25 deg, forecasts are updated every 6 hours, from 00 to 144 hours (6 days) at 3-hour increments for the four daily forecasts, plus 150 to 360 hours (15 days) at 6-hour increments for the 00z and 12z forecasts. Data is usually available by 8 hours after the Synoptic time (HH+8:00), so 08:00z, 20:00z (utc) for the 360-hour 00z and 12z forecasts, and 14:00z and 02:00z for the 06 & 18z forecasts.

In constrast to GFS (above), data is made available as a batch for each forecast, there is no "early" data available for shorter forecasts. A scheduled request (subscription) for 00z run should be requested for 8:00, etc. it wil usually be sent around 07:45.

A typical request might be:

```
send ECMWF:19N,35N,123W,102W|0.25,0.25|0,3..72|  
PRMSL,WIND
```

ECMWF's AIFS Model:

In addition to the physics-based IFS, ECMWF has also launched an operational version of their Artificial Intelligence Forecasting System, AIFS. This data is available from ECMWF's open-data project, see terms below.

Currently available parameters are:

- WIND (10 meters above surface)
- PRMSL (PResure/Mean Sea Level)
- TMP (temperature 2 meters above surface)

The grid is 0.25 x 0.25 deg, and forecasts are updated every 6 hours, from 0 to 360 hours forecast times (15 days) at 6-hour increments. Data is usually available 6 hours after the Synoptic time (HH+6), so 06:00z, 12:00z, 18:00z and 00:00z (utc) for the 00z, 06z, 12z and 18z model runs, repsecitvely. (Yes, earlier than ECMWF's IFS).

A typical request might be:

```
send AIFS:35N,19N,123W,102W|0.25,0.25|0,3..72|  
PRMSL,WIND
```

ECMWF terms: Data is copyright ECMWF and distribution is goverened by the Creative Commons CC-4.0-BY licence and the ECMWF Terms of Use - see <https://www.ecmwf.int/en/forecasts/datasets/open-data> for details.

ICON:

"ICON" is a global model run by Deutscher Wetterdienst (DWD), the German Met service. It is based on an Icosahedral (triangular) grid, hence the name, and converted to a lat-lon grid by Saildocs. It is a well-respected model, updated four

times a day, and available from Saildocs on a 1/8-deg grid (0.125 x 0.125), at 3-hour intervals out to 180 hours for the 00z and 12z model runs and 120 hours for the 06z and 18z runs. Data is generally available 4 hours after the synoptic time (e.g. 04z for the 00z run). Available parameters are wind, mean sea-level pressure and 500hPa height (WIND,MSLP,HGT500), also GUST, SFCTMP, AIRTMP (2m).

There is also a higher-resolution regional "nest" for Europe, bounded by the limits of 29.5N-70.5N and 023.5W-045.0E. The data grid is 0.0625 x 0.0625 deg (around 7km) available at 3-hour intervals out to 120 hours. Use the same "ICON" code, Saildocs will select the appropriate data for the requested latitude/longitude area.

A typical request could be:

```
send ICON:26N,20N,114W,105W|0.25,0.25|0,6..72|  
WIND,MSLP,HGT500
```

NAVGEN:

This is the US Navy model, formerly known as NOGAPS, with data available on a 1-deg grid with forecasts every 3 hours to 24, then 6 hours to 96, then 12 hours to 144, and updated every 6 hours by HH+06:00. Available parameters are PRMSL,WIND,HGT (the default is PRMSL,WIND). The general view is that GFS has better accuracy on average, but Navgem is an independent model and is valuable as a "second opinion".

At this time the US Navy server is not available for public access, and NAVGEM data is not currently available.

COAMPS:

This is a meso-scale (i.e. more detailed) regional model

provided by the US Navy. Data is available for three regions: eastern Pacific, west Atlantic, and Carib/Central America. The syntax is the same as for a basic grib, except the request-code "coamps:" is used in place of "gfs:" or "grib:".

For the four US coastal areas, forecast data is available on a 0.15 x 0.15-deg grid at 3-hour intervals out to 96 hours (except 72 hrs for w-Atl). Available (and default) parameters are WIND. The euro model is 96 hours at 6-hour intervals for pressure and wind.

The areas for the currently available coamps areas are:

- NE Pac: 34n,64n,172w,118w
- So Cal: 22n,43n,128w,109w
- Eq. Am: 12s,32n,122w,058w
- W Atl: 20n,55n,093w,055w
- Euro: 29n,66n,015w,040e

The same "coamps:" code is used for all regions, and Saildocs chooses the appropriate region based on the requested lat-lon area. Requests which fall partially outside the selected region will be trimmed at the edge of the region, and a zero-length file will result if the request is completely outside any of the regions. It is not possible to span multiple regions because data from different files does not match at the region edges, apparently a boundary-condition limitation with the model.

NOTE: The source for this data is the same as NAVGEM above, and is not currently available.

RTOFS:

This is an ocean current forecast model from NOAA's RTOFS system, based on the HYCOM model (see <http://polar.ncep.noaa.gov/ofs/> for more details). It was previously two different models and three overlapping areas (gulf-stream, Atlantic and Global) but now consolidated into a single global model available with the code "RTOFS". The resolution is 0.08

(1/12) deg.

RTOFS data is currently available out 72 hours at 3-hour increments generally available around 11:00z. A second model run provides forecasts beyond 72 hours, out to 192 hours (8 days) at 6-hour increments, generally available around 17:00z. If data beyond 72 hours is not yet available, then Saildocs will fill the request from the previously daily file.

Available RTOFS parameters are:

- CURRENT or CUR surface current)
- WTMP (water surface temperature)
- DSL_M sea-level deviation)
- SALTY (salinity)
- ICEC, (ice Concentration)
- ICETK (ice Thickness)

The default (if parameters are omitted) is CURRENT.

More information is available here: <https://polar.ncep.noaa.gov/global/about/>

OSCAR:

OSCAR is a 5-day historical current map in grib format, but is currently unavailable from Saildocs -- see <http://www.svsarana.com/oscar>

Gridded Marine Forecast Data

NDFD: NDFD (National Digital Forecast Database) is a US National Weather Service (NWS) project to make forecaster-generated weather forecast data available in gridded format for the NWS areas of responsibility. These data files are prepared along with the forecast charts (radiofax charts) and text bulletins. The advantage of the grib format is a much more efficient file format (and smaller files) than chart images.

NDFD is a developing project, and available data will be

expanded. Currently wind and wave data is available for the Pacific from 20S to the equator and west to 120W, equator to 30N and west to 140W, 30N to 49N and west to 133W, and offshore waters around Alaska and Hawaii. The current Atlantic coverage is 03N to 31N from the Gulf eastwards to Africa, and from 31N to 50N from the coast east to 064W.

To request NDFD data use the code "NDFD", specify the lat-lon area (respecting the available areas, above). The resolution is 0.12 deg (default is 1-deg if omitted), forecasts are available out to 7 days at 3 to 6-hour increments. Files are currently updated by 01:00z (for the 00z forecast), then 07:00, 13:00, 19:00 and 23:00z.

Currently available parameters are WIND,GUST,WAVES.

A typical request would be:

```
send NDFD:26N,20N,114W,105W|0.25,0.25|12,24..72
```

(Roughly Cabo to Puerto Vallarta, 0.25 deg grid, every 12 hours out to 3 days, about 12KB).

NDFD is "experimental" status which means that the data may not be available or correct (which is always true for computer data).

For more info about the NDFD program is here: https://ocean.weather.gov/opc_gridded_marine.php

To receive an updated copy of this document whenever changes are made, send the following one-line request to:

query@saildocs.com

```
subscribe gribmodels days=365
```

(The "days" parameter indicates the time that Saildocs will check for changes: "365" means one year, "0" means indefinitely).

To get general info about Saildocs send a (blank) email to:

info@saildocs.com (auto-responder)

To see the real-time status of grib data available from Saildocs, see this page: <https://saildocs.com/gribstatus>

For problems/questions relating to Saildocs, send an email to: support@saildocs.com

Thanks for using Saildocs and Good Sailing!

Updated 2025-12-14: NOAA's AIGFS, GFS tweaks